Question 1: Explain the difference between List and Tuple.Also write a Python program to:

```
•   Create a list
•   Convert it into a tuple
•   Try modifying both
```

```
# list are mutable,slower in performance and dynamic data while tuple are immutable , faster and fixed data
my_list = [1,2,3,4,5] # Creating list
my_tuple = tuple(my_list) # Converting to tuple
type(my_tuple)
```

```
tuple
```

```
my_list[2]= 4 # changing index two value with 4. List so mutable
my_list
```

```
[1, 2, 4, 4, 5]
```

```
my_tuple[2]=4 # trying to change index two value with 4. Error occor as tuple are immutable.
my_tuple
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_6358/276107073.py in <cell line: 0>()
----> 1 my_tuple[2]=4
      2 my_tuple

TypeError: 'tuple' object does not support item assignment
```

Next steps: Explain error

Question 2: What is a Set?Explain how sets remove duplicates.Write a program to remove duplicates from a list and print unique values.

```
# Set is unordered collection of unique elements.
# Set removes duplicates using hashing. Eash element is stored only if it has unique hash value otherwise ignored it.

my_list1 = [1,2,2,3,4,4,5]
unique = set(my_list1)     # only gives unique values only
print(unique)
```

```
{1, 2, 3, 4, 5}
```

Question 3: Write a program to:

- Reverse a list using slicing
- Also print original list to show it is unchanged

```
my_list2 = [1,2,3,4,5]
print(my_list2[::-1])  # it show reverse list
print(my_list2)      #it will show the original list
```

```
[5, 4, 3, 2, 1]
[1, 2, 3, 4, 5]
```

Question 4: Write a program to:

- Create a dictionary of student details
- Access the name
- Remove age using pop()
- Print length of dictionary

```
student = {                   #Creating a dict student
    "name":"Matrika",
    "age":"33"
}
print(student)
```

```
{'name': 'Matrika', 'age': '33'}
```

```
student.pop("age")         # Remove age using pop
```

```
'33'
```

```
print(student)       # no age in student dict
```

```
{'name': 'Matrika'}
```

```
len(student)      # len of dictionary
```

```
1
```

Question 5: Write a program to:

- Take a string
- Check if it contains only alphabets
- Print "Valid" or "Invalid"

```
str1 = "Matrika"

if str1.isalpha():     # check if all character are alphabets or not
  print("Valid")
else:
  print("Invalid")
```

```
Valid
```